

INDIAN STATISTICAL INSTITUTE

Mathematics-3 (BSDS)

QUIZ - 3 Examination

DATE: 1/12/2025

Total Marks: 10

Time: 3:30 – 4:15 pm

MARKS WILL BE AWARDED ONLY FOR THOSE ANSWERS WITH PROPER
JUSTIFICATION

Question 1: Let (X, d) be a metric space. Define a function $d' : X \times X \rightarrow \mathbb{R}$ by

$$d'(x, y) = \frac{d(x, y)}{1 + d(x, y)}.$$

Prove that (X, d') is a metric space. If (X, d) is complete, show that (X, d') is complete [6]

Question 2: Prove that every closed subspace of a compact space is compact. [4]

—ALL THE BEST—